REVATHY VENUGOPAL

R&D Engineer • Python, APIs & Applied AI

Lyon, France • revathyvenugopal162@gmail.com • Revathyvenugopal162
linkedin.com/in/revathy-venugopal • revathyvenugopal162.github.io



Python engineer and open-source maintainer with four years of experience building developer-facing APIs, applied-AI systems and large documentation platforms for engineering software. Focused on LLM and RAG pipelines, agentic workflows, MCP tooling and open-source automation. Speaker at SciPy and EuroSciPy.

Experience

R&D Engineer, Core Python Team — **May 2022 – Present**

Ansys • Lyon, France

- **Applied AI.** Designed RAG pipelines and agentic LLM workflows for engineering applications, covering document chunking, embeddings, vector retrieval, hybrid semantic and keyword search, and response evaluation; exposed internal capabilities to AI assistants through Model Context Protocol (MCP) servers.
- **Backend services.** Built a multi-tenant API service with dynamic index loading, giving each tenant an isolated and horizontally scalable deployment.
- **Search platform.** Re-architected documentation search from a Python scraper with a MeiliSearch backend to a build-time static index served by Fuse.js, removing authentication, VPN and multi-version re-indexing constraints.
- **Developer automation.** Built a GitHub CLA bot that automates contributor licence agreement signing, verification and pull-request status checks across the organisation's public repositories.
- **Libraries and tooling.** Ship Python packages, project templates, reusable CI/CD actions and release automation used across the ecosystem, plus the shared documentation theme and API-reference templates.
- **Ownership.** Maintain a public Python simulation library end to end: API design, code review, releases and community support; lead design reviews and mentor new contributors.

Machine Learning & Robotics Intern — **Sep 2021 – Mar 2022**

SITIA Robotique • Nantes, France

- Built Python computer-vision software for obstacle detection and classification using TensorFlow and OpenCV.
- Implemented real-time perception and Kalman-filter-based speed control for autonomous agricultural robots.

Open source

- Core team of a large open-source Python ecosystem: shared libraries, project templates, CI/CD actions and release automation.
- Maintainer of a public simulation library: feature development, issue triage, documentation and releases.
- Author of a documentation theme with a built-in static search engine and automated API-reference templates, and of a Python indexer that makes MeiliSearch practical for multi-version documentation.

Speaking

Sphinx Themes for Documentation • watch — EuroSciPy 2025, Kraków
AutoAPI Templates with Sphinx Themes — SciPy 2024, Tacoma
PyMeiliSearch, lightning talk • watch — SciPy 2023, Austin

Skills

AI & ML	LLMs • RAG • agentic workflows • MCP • embeddings • vector stores • prompt engineering • TensorFlow • PyTorch • OpenCV
Programming	Python • JavaScript • C • HTML • CSS
Backend & web	FastAPI • Flask • REST APIs • asyncio • pytest • Fuse.js • MeiliSearch
Platform	Git • GitHub Actions • CI/CD • Docker • Sphinx • LaTeX
Languages	English (C1) • French (B1)

Education & certifications

MSc, Embedded Systems • ESIGELEC, Rouen — 2020 – 2022
MSc, Automotive Embedded Systems • Manipal School of Information Sciences — 2020 – 2021
BTech, Electrical & Electronics Engineering • College of Engineering, Munnar — 2016 – 2020
Embeddings & vector spaces in RAGs • Docker and Kubernetes • *Pluralsight* — 2023 – 2024